Sustainable Industrialization and Resource-led Growth

Eastern India sits on a paradox. West Bengal, Odisha and Jharkhand hold much of the country's coal, iron ore and bauxite, and the region built India's first great industrial belt around Jamshedpur, Durgapur, Rourkela and the Damodar valley. Those towns showed the nation how to make steel. Yet drive through the mining districts today and the wealth underground is hard to see above it. Economists have a name for this pattern: the resource curse. My generation's question is how to break it, and how to turn natural resources into prosperity that lasts without destroying the rivers, forests and climate that prosperity depends on.

The first lesson from the past century is plain: digging is not the same as developing. For decades, coal and ore left the region as raw material while the ash ponds and mine voids stayed behind. The Raniganj and Jharia coalfields powered the railways and factories of independent India. They also left land that sinks underfoot and fires that have burned underground for decades. A better strategy starts closer to the resource: more processing, more manufacturing, more of the value staying where the mineral came out of the ground.

Beyond these old lessons, the world economy has shifted. India has promised net zero emissions by 2070 and aimed for 500 gigawatts of non-fossil electricity capacity by 2030; the Press Information Bureau reported that the country crossed 50 percent non-fossil installed capacity in 2025, five years ahead of schedule. Buyers abroad increasingly ask for green steel and low-carbon aluminium. For a region built on coal, this can look like a threat. I see it as the region's biggest opening, if we move early. Odisha and Jharkhand get strong sunshine, and the existing mines, railways and ports, including Haldia, give the region the base to build what the energy transition needs: electrolysers, transmission equipment, electric vehicle parts and, eventually, steel made with green hydrogen instead of coking coal. I will not pretend this is cheap. Green hydrogen costs more than the fuels it replaces today, and pretending otherwise would weaken every argument above it. Costs can fall as production scales, so investing early could help the region compete.

Cleaner growth also means wasting less of what we already dig up. This is not theory. In Surat, CSIR's Central Road Research Institute and the Ministry of Steel built India's first road from processed steel slag, the waste that steel plants pile up. Fly ash, which eastern power plants produce in huge quantities, goes into bricks instead of ponds when there is a buyer for it. Old mine voids can store water or carry floating solar panels. Industrial towns across the region can copy this pattern: one factory's waste becoming another's raw material.

No transition works if it abandons the people who live where the mines are. Lakhs of coalfield families depend on mining wages, and Adivasi communities depend on forests that mining disturbs. India already has a tool for this: District Mineral Foundations, which move a share of mining royalties back into the affected districts. Where they are used well, they pay for water, health and retraining. The region should insist they are used well everywhere. On one hard question I will take a side: new mining should be approved only when the restoration fund and a dated mine-closure plan exist before the first truck rolls.

Restoration belongs in the project's first budget, not its last apology.

The ecological balance is not decoration on top of all this. The Sundarbans, the world's largest mangrove forest, shields Kolkata and the Bengal coast from cyclones. The forests of Jharkhand and Odisha feed the rivers the factories drink from. Every project should answer one plain question: does it leave the region's natural capital stronger or weaker? Pollution monitoring that any citizen can check would make that question practical instead of rhetorical.

My proposals are three. First, build green industrial corridors along the existing mineral belt, so renewable power, research centres and skill institutions grow where the factories already are. Second, use the region's universities and technical institutes, including those in West Bengal, to train the solar technicians and environmental engineers the transition needs, so local youth fill these jobs. Third, and most important, publish the data: mine-by-mine restoration accounts, river and air quality dashboards, open to everyone. Eastern India has always measured growth in tonnes. It should start counting jobs kept, rivers cleaned and land restored.

Eastern India answered one question for India: it showed the nation how to industrialise. The harder question is how to keep growing without exhausting the land and the people who make growth possible. Add value at home, move early on clean energy, waste less, protect workers and guard the rivers and forests, and the minerals become what they should have been all along: the base of a prosperity that lasts. The coal seam lit India's first factories; the same soil, handled wisely, can power its greenest ones.